Composition and action normalization: reciprocal mechanics, rotations and spin phases

Interaction can constrain relative normalizations while leaving an absolute action coefficient free. In the canonical product algebra the decisive compatibility identity is

$$0 = D[p_{ia}, p_{jb}] = i(h_i - h_j)\partial_{ia}\partial_{jb}V.$$

Thus a Newton derivation and a nonzero mixed force derivative force equal constituent constants on each connected interaction component. The common zero branch remains admissible. Mechanical angular closure gives a separate normalization test under faithful constituent observables; spatial covariance alone permits unequal constants.

The hierarchy begins with reciprocal spring calibration and ends with the extra phase premise on a spin sphere. Reciprocal forces fix inertia ratios. A supplied phase constant $K > 0$ makes global spin patching require $2s/K \in \mathbb{Z}$. Neither the classical model, the algebraic equality nor the phase ratio selects an absolute nonzero physical $\hbar$. Classical fluctuation composition has its own probabilistic hypotheses and remains in its separate treatment.

These are conditional written proofs and established geometric comparisons, not independent peer review or a claim of literature novelty. In the algebraic sections h denotes a reduced phase constant: quantum identification would mean $h = \hbar$ and $h_P = 2\pi\hbar$. K in the phase section is supplied, and equation labels M, A and P distinguish the three premise sets.

Reciprocal mechanical calibration

Measured response and reciprocal completion

Let two coordinates $x, y \in \mathbb{R}$ have the same length units and share physical time t. Suppose their measured equations are

$$\ddot{x} = -\alpha(x - y), \qquad \ddot{y} = \beta(x - y), \qquad \alpha, \beta > 0. \tag{M1}$$

Both coefficients have units time^{-2}. Isolated free motion would leave the sectors' positive inertias m_1, m_2 independently arbitrary. Now require the specific mechanical completion

$$L = \tfrac{1}{2}m_1\dot{x}^2 + \tfrac{1}{2}m_2\dot{y}^2 - \tfrac{1}{2}k(x - y)^2, \qquad k > 0. \tag{M2}$$

Here masses have mass units and k has mass/time^2 units. This assumes a diagonal constant kinetic energy and one reciprocal position-dependent interaction. It is a physical model premise, not a consequence of observing arbitrary coupled motion. Its forces are $F_1 = -k(x - y)$ and $F_2 = k(x - y)$. The Euler–Lagrange equations agree with (M1) exactly when

$$m_1\alpha = m_2\beta = k, \qquad \frac{m_2}{m_1} = \frac{\alpha}{\beta}. \tag{M3}$$

Thus fixed responses admit precisely the positive family $(m_1, m_2, k) = m_1(1, \alpha/\beta, \alpha)$, with the entries interpreted in their stated units. Coupling constrains relative normalization; equal inertias follow only if the two acceleration responses are equal.

Exchange currents and common scaling

Writing $r = x - y$, the conserved energy and total momentum are

$$H = \tfrac{1}{2}m_1\dot{x}^2 + \tfrac{1}{2}m_2\dot{y}^2 + \tfrac{1}{2}kr^2, \qquad P = m_1\dot{x} + m_2\dot{y}. \tag{M4}$$

Indeed $\dot{T}_1 = -kr\dot{x}$, $\dot{T}_2 = kr\dot{y}$, and $\dot{V} = kr(\dot{x} - \dot{y})$, so $\dot{H} = 0$; the two forces give $\dot{P} = 0$. To display an exchange current, assign half the interaction energy to each sector: $E_1 = T_1 + V/2$, $E_2 = T_2 + V/2$. Then

$$\dot{E}_1 = -\tfrac{k}{2} r(\dot{x} + \dot{y}) = -\dot{E}_2. \tag{M5}$$

The half split is an accounting convention; conservation of H is independent of it. Generic initial data give nonzero exchange. Measuring power in an independently calibrated energy unit would add information beyond (M1).

For every dimensionless $\eta > 0$, replace (m_1, m_2, k) by $(\eta m_1, \eta m_2, \eta k)$. Equations (M1), all coordinate histories with the same initial positions and velocities, and their periods remain unchanged. L, H, P, E_i and the currents in (M5) scale by η. So does the trajectory action $S = \int L\, dt$ on any fixed time interval. Independent rescaling of only one sector generally violates (M3); common rescaling does not.

An on-shell positive action diagnostic makes this freedom visible without cancellation in S. Set $M = m_1 + m_2$, $\mu = m_1 m_2/M$ and $X = (m_1 x + m_2 y)/M$. Then

$$L = \tfrac{1}{2} M \dot{X}^2 + \tfrac{1}{2} \mu \dot{r}^2 - \tfrac{1}{2} k r^2, \qquad \omega^2 = k/\mu = \alpha + \beta. \tag{M6}$$

In the stationary centre frame choose $r = A\cos(\omega t)$, $A > 0$. With $p_r = \mu \dot{r}$, its closed-orbit action is

$$I = \frac{1}{2\pi} \oint p_r\, dr = \frac{1}{2} \mu \omega A^2 = \frac{E_{\mathrm{rel}}}{\omega}. \tag{M7}$$

The integral follows from $p_r \dot{r} = \mu A^2 \omega^2 \sin^2(\omega t)$ over one period. It has action units. Common rescaling sends I to ηI at fixed trajectory. Even for fixed masses and spring, the admitted small amplitudes $A \downarrow 0$ send I to zero. The full Lagrangian action over this stationary-centre period is zero because the kinetic and potential integrals coincide; (M7) must not be identified with that signed integral.

The relative frequency is strictly positive for this fixed two-body model. The full system also has a free centre coordinate, so it has a zero-frequency mode. This is a finite-dimensional classical frequency statement, with no volume or continuum claim and no quantum energy gap.

Connected networks and the cycle test

The relative mechanism extends to a finite connected undirected graph of scalar coordinates, all measured in the same length and time units. On every edge $\{i, j\}$ suppose both directed responses $a_{ij}, a_{ji} > 0$ are known:

$$\ddot{q}_i = -\sum_{j \sim i} a_{ij}(q_i - q_j). \tag{M8}$$

Seek only the mechanical class

$$L = \tfrac{1}{2} \sum_i m_i \dot{q}_i^2 - \tfrac{1}{2} \sum_{\{i,j\}} k_{ij}(q_i - q_j)^2, \qquad m_i > 0, \quad k_{ij} = k_{ji} > 0. \tag{M9}$$

Comparison of coefficients requires $m_i a_{ij} = m_j a_{ji} = k_{ij}$. Hence the necessary cycle condition is

$$\prod_{\ell=0}^{s-1} \frac{a_{i_\ell i_{\ell+1}}}{a_{i_{\ell+1} i_\ell}} = 1, \qquad i_s = i_0. \tag{M10}$$

It is sufficient as well: choose any $m_0 > 0$, transport masses along a spanning tree using $m_j/m_i = a_{ij}/a_{ji}$, and apply (M10) on each remaining edge's fundamental cycle. This makes its transported ratio agree with the edge ratio. All masses are positive and define the required springs. Any two solutions have the same mass ratio along every edge; connectivity makes

their quotient one common constant. A disconnected graph instead retains one constant per connected component.

For a concrete failed test take a triangle with $a_{12} = 2\gamma$ and all other directed edge coefficients $\gamma > 0$. The cycle $1 \to 2 \to 3 \to 1$ has product 2. The required ratios would give $m_2 = 2m_1$, $m_3 = m_2$, and $m_1 = m_3$, a contradiction. Such responses cannot have completion (M9). This rejects that diagonal-inertia reciprocal-spring model; it does not exclude completions with additional variables or a different kinetic form.

Constituent algebras and the meaning of rotations

Take finitely many labelled bodies, $i = 1, \ldots, N$, with masses $m_i > 0$, three spatial coordinates and three momenta. Assume an associative unital complex algebra with self-adjoint generators satisfying

$$[q_{ia}, q_{jb}] = [p_{ia}, p_{jb}] = 0, \qquad [q_{ia}, p_{jb}] = i\delta_{ij}\delta_{ab}h_i 1, \quad a, b \in \{x, y, z\}. \tag{A1}$$

The scalars $h_i \in \mathbb{R}$ remain fixed when the body is put in a composite. Distinct constituent algebras commute. Assume also that the normally ordered monomials $q^\alpha p^\beta$ are linearly independent: we retain the full tensor product of canonical polynomial algebras, including its commutative factors when some $h_i = 0$. This is the precise independence used in coefficient comparisons. Statistical independence of states is unnecessary; entangled states are allowed. For unbounded operator realizations all displayed identities are on a common invariant dense domain representing these polynomials faithfully.

Define, with $\epsilon_{xyz} = 1$,

$$L_{ia} = \sum_{b,c} \epsilon_{abc} q_{ib} p_{ic}, \qquad L_a = \sum_i L_{ia}. \tag{A2}$$

Each term in (A2) has unambiguous ordering because q_{ib} and p_{ic} commute for $b \neq c$. The additivity and mechanical normalization in (A2) are hypotheses. Independently rescaling each body's momentum would change its measured quantity of motion and these hypotheses.

Three distinct requirements can now be stated precisely.

- **Spatial covariance:** the map $q_i \mapsto Rq_i$, $p_i \mapsto Rp_i$ is an algebra automorphism for every $R \in SO(A3)$.
- **Mechanical closure:** for one real h, $[L_a, L_b] = ih \sum_c \epsilon_{abc} L_c$ on the full algebra.
- **Mechanical generator identification:** for $h \neq 0$, the generator $K_a = L_a/h$ satisfies $[K_a, q_{ib}] = i \sum_c \epsilon_{abc} q_{ic}$ and $[K_a, p_{ib}] = i \sum_c \epsilon_{abc} p_{ic}$ for every body.

The last convention gives $U(\theta) = \exp(-i\theta K_a)$ and $U(\theta)^\dagger q_{ib} U(\theta) = q_{ib} - \theta \sum_c \epsilon_{abc} q_{ic} + O(\theta^2)$, the usual active rotation of the vector's components about axis a. Integration to a group representation additionally needs a strongly continuous unitary action with these generators. Lie commutator identities alone leave those domain and integrability obligations open. For $h \neq 0$, such an action is a representation of $SU(A2)$; descent to $SO(A3)$ additionally requires a full 2π rotation to act as the identity.

The orbit-phase premise $e^{iL\theta/h_i}$ yields (A1) only after an additional canonical observable/quantization prescription. A single orbit's angular descent condition by itself supplies neither the tensor algebra nor its mechanical generator identification. The present theorem begins at that explicit algebraic extension.

Mechanical closure forces a shared constant

Theorem 1. Under (A1)–(A2) and the monomial independence hypothesis, mechanical closure holds for a scalar h if and only if $h_i = h$ for all i. It suffices to impose the single identity

$[L_x, L_y] = ihL_z$. For $h \neq 0$, mechanical generator identification is also equivalent to $h_i = h$ for all i.

Proof. The product rule for commutators gives

$$[L_{ia}, q_{jb}] = i\delta_{ij} h_i \sum_c \epsilon_{abc} q_{ic}, \qquad [L_{ia}, p_{jb}] = i\delta_{ij} h_i \sum_c \epsilon_{abc} p_{ic}. \tag{A3}$$

For example, writing $L_{ix} = q_{iy}p_{iz} - q_{iz}p_{iy}$ and $L_{iy} = q_{iz}p_{ix} - q_{ix}p_{iz}$, their commutator has exactly the surviving terms

$$-ih_i q_{iy} p_{ix} + ih_i q_{ix} p_{iy} = ih_i L_{iz}.$$

Different bodies commute, so closure of the sum is equivalent to

$$\sum_i (h_i - h) L_{iz} = 0. \tag{A4}$$

The monomial $q_{ix}p_{iy}$ in (A4) occurs only in L_{iz}, with coefficient $h_i - h$. Independence makes that coefficient zero for each i, including a constituent with $h_i = 0$. Conversely, equal constants give closure in all three directions by the same calculation.

For generator identification, (A3) gives, for example, $[L_x, q_{iy}] = ih_i q_{iz}$, whereas the required action is ihq_{iz}. Independence makes $q_{iz} \neq 0$, hence $h_i = h$. The converse follows directly from (A3). □

A restricted observable space needs the corresponding separation hypothesis in place of monomial independence: each relation $\sum_i c_i L_{ia} = 0$ must force every $c_i = 0$. Compression to a subspace where all orbital L_i vanish makes closure vacuous for arbitrary h_i. A single orbit, expectation value or rigidly constrained configuration can likewise lose the necessary separation. The theorem uses observable identities valid for all admitted preparations and configurations, with the constituents still individually accessible.

Transitivity. Form a graph whose vertices are bodies or species with persistent constants h_i. Put an edge between two vertices when their full pair algebra is admitted and its mechanical sum is required to have one closure constant. Theorem 1 gives equality on each edge and therefore throughout each connected component. Connectedness gives one constant for the whole class, without assuming that every pair interacts directly. Disconnected components remain unrelated unless a common composite or generator condition links them. A context-dependent h_i would require a further preparation-equivalence premise before this argument could compare contexts.

Role of three dimensions. A single planar rotation generator has $[L_z, L_z] = 0$, which tests no normalization. In three dimensions the non-abelian bracket in (A4) compares the mechanical sums internally. Requiring the correct action on each q_i, p_i already compares normalizations in two dimensions by (A3). Three dimensions strengthen the closure test; generator identification itself needs only one nontrivial rotation plane.

Rotation covariance and interaction compatibility

Covariance with unequal constants

For arbitrary h_i, each scalar canonical block in (A1) is rotation invariant. An explicit associative realization on polynomials is

$$f * g = \mu \exp\left[\frac{i}{2} \sum_i h_i P_i\right] (f \otimes g), \qquad P_i = \sum_a \left(\partial_{q_{ia}} \otimes \partial_{p_{ia}} - \partial_{p_{ia}} \otimes \partial_{q_{ia}}\right). \tag{A5}$$

The exponential terminates on polynomials. The constant derivative operators commute, so either association of three factors gives the same exponential of the sum over their three pairings. Each P_i is invariant under simultaneous spatial rotations. Thus (A5) supplies covariance for every tuple $(h_1, \ldots, h_N)$.

If all $h_i \neq 0$, define

$$K_a = \sum_i L_{ia}/h_i.$$

Equations (A3) give $[K_a, K_b] = i\epsilon_{abc}K_c$ (summed c) and the correct rotations of all constituent observables. Concretely, on $L^2(\mathbb{R}^{3N})$, $p_i = -ih_i\nabla_i$ yields $\mathbf{K} = -i\sum_i \mathbf{q}_i \times \nabla_i$, which integrates to simultaneous scalar rotations for arbitrary nonzero h_i. The identification $K_a = L_a/h$ adds exactly the normalization tested in Theorem 1. With some zero factors, spatial covariance still holds as an automorphism; commutators with their central observables vanish, so their rotations require an outer derivation.

Central interactions and conservation

For classical Newton dynamics about a fixed inertial origin, $\dot{q}_i = p_i/m_i$ and $\dot{p}_i = \sum_{j\neq i} F_{ij}$, the strong central form of the Third Law is $F_{ji} = -F_{ij}$ with F_{ij} parallel to $q_i - q_j$. It gives

$$\dot{\mathbf{L}} = \sum_{i<j}(q_i - q_j) \times F_{ij} = 0. \tag{A6}$$

Equal and opposite forces together with centrality do the work here. Equation (A6) uses neither a commutator nor an action constant. Transferring this conservation law to the deformed observable algebra needs a specified evolution law.

Theorem 2 (conservation propagates equality along interacting pairs). Assume (A1), faithful configuration observables on open sets, and a Hamiltonian

$$H = \sum_i \frac{p_i^2}{2m_i} + \sum_{i<j} V_{ij}(r_{ij}), \qquad r_{ij} = |q_i - q_j|. \tag{A7}$$

Assume commutator differentiation $[p_{ia}, V] = -ih_i\partial_{q_{ia}}V$ on a common domain. This holds algebraically for polynomial potentials, or on a smooth domain away from collisions for the coordinate realization. Give time evolution one fixed nonzero action normalization h_t, $\dot{A} = [A, H]/(ih_t)$. Require conservation of the mechanical sum for every admitted pair interaction, with other pair couplings independently switchable off. For each tested edge require $V'_{ij}(r_{ij}) \neq 0$ on some open set allowing $q_i \times q_j \neq 0$. Then conservation forces $h_i = h_j$ on that edge, and therefore one value per connected interaction component.

Proof. The kinetic energies commute with each L_i. For one pair, $\nabla_j V_{ij} = -\nabla_i V_{ij}$, and centrality gives $(q_i - q_j) \times \nabla_i V_{ij} = 0$. Consequently

$$[\mathbf{L}, V_{ij}] = i(h_i - h_j)q_i \times \nabla_i V_{ij} = +i(h_i - h_j)\frac{V'_{ij}(r_{ij})}{r_{ij}} q_i \times q_j. \tag{A8}$$

The configuration factor is a nonzero observable on the stated open set. Conservation, equivalent to $[\mathbf{L}, H] = 0$, forces $h_i = h_j$. When constants agree on every interacting edge, (A8) proves the converse conservation statement as well. □

One central harmonic interaction suffices as a polynomial test: $V_{ij} = k_{ij}|q_i - q_j|^2/2$, $k_{ij} > 0$, gives the exact defect $+ik_{ij}(h_i - h_j)q_i \times q_j$ (the displayed sign includes the earlier correction). This permits full configuration variation and keeps every step inside the polynomial algebra. Restriction to collinear configurations or to a pair's fixed centre-of-mass frame would remove this particular torque test; conservation about arbitrary fixed origins restores its stated scope. An external central potential for each isolated body gives no such exchange test and allows different constants.

Theorem 2 compares the h_i but leaves their common value relative to h_t undetermined. Demanding the full Newton equations with the physical Hamiltonian (A7) gives the stronger identities

$$\dot{q}_i = \frac{h_i}{h_t}\frac{p_i}{m_i}, \qquad \dot{p}_i = -\frac{h_i}{h_t}\nabla_i V. \tag{A9}$$

Requiring $\dot{q}_i = p_i/m_i$ for unrestricted momenta forces $h_i = h_t$. This route already assumes a nonzero inner-commutator clock. Alternatively, assigning each body its own denominator and retaining all Newton equations must respect the shared product: for any derivation D with $Dp_{ia} = -\partial_{ia}V$,

$$0 = D[p_{ia}, p_{jb}] = i(h_i - h_j)\partial_{ia}\partial_{jb}V, \qquad i \neq j. \tag{A10}$$

A nonzero mixed force derivative again equates the constants. This is the main dynamical result of the note under the displayed derivation premise: it needs no centrality and no angular momentum, and it holds for every interacting pair. Equation (A10) follows by applying the Leibniz rule to the zero cross commutator; it exposes the compatibility obligation of separate denominators. The common zero branch satisfies (A10) and supports ordinary classical Hamiltonian evolution by a Poisson derivation.

The common zero branch and angular representations

Corollary (universality and positivity). Theorem 1 forces a common real value, with $h = 0$ allowed. If one constituent has an independently established $h_i \neq 0$, connected composition propagates its nonzero value to the entire component. The positivity input lies in that reference constituent. Theorem 2 likewise permits all $h_i = 0$; its commutator clock then gives trivial evolution. Classical nontrivial dynamics uses the Poisson bracket instead.

For the common family (A5), all real h give associative algebras. At $h = 0$ the product is ordinary multiplication. Spatial rotations remain non-abelian: their derivations are $D_a f = \{f, L_a\}_{\mathrm{P}}$, and classical angular momenta obey $\{L_a, L_b\}_{\mathrm{P}} = \epsilon_{abc}L_c$. The Poisson algebra supplies the rotation structure while the multiplication algebra is commutative. For polynomial observables the common Moyal family has

$$\lim_{h\to 0} \frac{[f, g]_{*_h}}{ih} = \{f, g\}_{\mathrm{P}}. \tag{A11}$$

Requiring a nontrivial rotation to be implemented specifically by $\exp(-i\theta L_a/h)$ selects $h \neq 0$ in the premise. Its formula requires division by h; classical rotations remain perfectly defined by their derivations at zero. A strongly continuous unitary rotation representation alone also leaves this choice open: pullback rotates classical phase-space functions in L^2 using differential generators, while measured classical angular momenta act by multiplication.

For $h \neq 0$, supply self-adjoint angular generators integrating to a unitary $SU(A2)$ representation. Its irreducible sectors have

$$J^2 = h^2 j(j+1), \qquad J_z = hm, \qquad j \in \{0, \tfrac{1}{2}, 1, \tfrac{3}{2}, \ldots\}, \quad m = -j, -j+1, \ldots, j. \tag{A12}$$

Here J denotes total angular momentum, including any supplied internal spin. To recall the dimensionless argument, put $K = J/h$ and $K_\pm = K_x \pm iK_y$. Then $[K_z, K_\pm] = \pm K_\pm$ and $K_\mp K_\pm = K^2 - K_z^2 \mp K_z$. In an irreducible unitary sector, positivity bounds the ladder above and below. Its highest weight j has $K^2 = j(j+1)$; the lowest weight is $-j$, so $2j$ is a nonnegative integer. A full rotation acts by $e^{-2\pi i m} = (-1)^{2j}$; $SO(A3)$ descent retains integer j.

For ordinary scalar orbital wavefunctions, $L = q \times (-ih\nabla)$ has integer orbital labels and spherical harmonics. The canonical angular algebra and this orbital spectrum are standard; see Tong, *Quantum Mechanics*, §§4.1.1–4.1.2 (passage, commutator and spherical-harmonic derivations read). Half-integer spin requires additional internal degrees of freedom or an explicitly

different state space. The central-force orbital construction supplies no such degrees of freedom by itself. If independent spin observables commute with the orbital variables, requiring $J = L + S$ to rotate those variables still gives (A3), hence the same orbital universality result. Requiring it also to rotate each nontrivial spin algebra fixes the spin normalization in the same way.

Equation (A12) supplies dimensionless spectral ratios. The Casimir is discrete; its factor $j(j+1)$ is integer for integer j and can be fractional for half-integer j. Conditional on fixed $|h| > 0$, the smallest nonzero J^2 among all integer sectors is $2h^2$, and among all $SU(A2)$ sectors it is $3h^2/4$. A chosen representation can omit these sectors; the scalar sector $j = 0$ always remains an allowed representation. Both spectral factors leave the dimensional value of h free.

More explicitly, rescaling $J \mapsto \lambda J$, $h \mapsto \lambda h$ for $\lambda > 0$ preserves the generators K, all rotation matrices, and the integer/half-integer labels. A measured nonzero magnitude ℓ and a known $j > 0$ would fix $|h| = \ell/\sqrt{j(j+1)}$, using that dimensional measurement as input. With only ℓ fixed, unbounded j allows values tending to zero. Several specified component and Casimir eigenvalues can further constrain j and their ratios; this constitutes additional spectral data. The scale freedom of the representation conditions alone remains.

Relation to fluctuation composition and dimensional selection

The classical composition note, §§1–3 (full-read) concerns a fluctuation coefficient $\kappa(m) \geq 0$ and independent displacement variances. Its whole-body premise is

$$\kappa(m_1 + m_2) = \frac{m_1\kappa(m_1) + m_2\kappa(m_2)}{m_1 + m_2}.$$

For all positive masses, $f(m) = m\kappa(m)$ is additive and nonnegative, hence linear; $\kappa(m) = K \geq 0$. A reference constituent then supplies $K = m_0 u_0^2/\lambda_0 > 0$ under the stated nonzero-speed and finite-rate premises. The rotation theorem replaces this mass-additivity premise with mechanical generator identification and separation of constituent observables. A connected species graph suffices, and arbitrary positive masses need not be admitted. Both arguments propagate a coefficient; each obtains positivity only with additional input. Identifying K with the phase constant h remains another physical obligation.

There is an exact parallel in centre and relative variables. Set $M = m_1 + m_2$, $\mu = m_1 m_2/M$,

$$R = \frac{m_1 q_1 + m_2 q_2}{M}, \quad r = q_1 - q_2, \quad P = p_1 + p_2, \quad p = \frac{m_2 p_1 - m_1 p_2}{M}.$$

Bilinearity in (A1) gives

$$\begin{aligned} [R_a, P_b] &= i\delta_{ab}\frac{m_1 h_1 + m_2 h_2}{M}, \quad & [r_a, p_b] &= i\delta_{ab}\frac{m_2 h_1 + m_1 h_2}{M}, \\ [R_a, p_b] &= i\delta_{ab}\frac{\mu}{M}(h_1 - h_2), & [r_a, P_b] &= i\delta_{ab}(h_1 - h_2). \end{aligned} \tag{A13}$$

Thus canonical independence of the centre and relative algebras forces $h_1 = h_2$. The classical note has the corresponding covariance $\mathrm{Cov}(\Delta R, \Delta r) = \Delta(\kappa_1 - \kappa_2)/M$; for its Gaussian preparations vanishing covariance gives statistical independence. Equation (A13) concerns commuting observable algebras, which can carry correlated or entangled states. The two independence premises have distinct meanings despite their matching coefficients.

The fifth-postulate note (full-read) places these results as follows. Its Theorem B obtains a single Moyal constant under full affine symplectic covariance. Applying that premise to the combined phase space already includes transformations mixing bodies, including (A13), whereas separate one-body covariance allows one constant per factor as in (A5). Theorem 1 isolates a narrower comparison using physical angular generators; it leaves general star-product uniqueness to the stronger covariance theorem. Theorem A retains the common zero and nonzero families

under its Newton equations. Theorems C and E give conditional disturbance and concentration bounds once $|h| > 0$ is supplied. Theorem D records action rescaling. The Gaussian state restriction of B' supplies its own ζ in a commutative algebra; spatial rotations and the present commutator test leave a relation $2\zeta = |h|$ to an additional premise.

Finally, the dimensional note, Theorem A (full-read) applies to a universal floor $g(c)$ determined by fixed constants c_r. If an action product exists, its form is

$$g(c) = \left(\prod_r c_r^{a_r} \right) F(\pi_1, \ldots, \pi_s), \qquad \sum_r a_r d_r = (1, 2, -1). \tag{A14}$$

It reduces to a pure number times the product when the fixed constants have no independent dimensionless products. Rotation angles, group structure constants, j and m are dimensionless and add no action unit to (A14). When the fixed constants admit no action-dimensional product, Theorem A allows only a zero or infinite universal floor; dimensionless rotations preserve that conclusion. Treating h as a new independent fixed constant makes $|h|$ the supplied unit. Deriving a universal $|h|$ from pre-existing constants instead requires dimensional homogeneity and the same covariance argument as (A14); angular universality provides no value for its dimensionless factor or proof of its positivity. For the fixed $k_e = e^2/(4\pi\epsilon_0)$, c and G of that note, this would read $|h| = Ck_e/c$. The rotation theorem leaves C undetermined. A coefficient and the infimum of a specified action observable also need a proved relation before (A14) can be used as a floor statement.

Global spin phases

Classical patches and complete dynamics

Let the spin direction n lie on the unit sphere with polar coordinates (θ, ϕ). Supply $s > 0$ with action units and define

$$\omega_s = s \sin\theta \, d\theta \wedge d\phi, \qquad \int_{S^2} \omega_s = 4\pi s. \tag{P1}$$

For any smooth real Hamiltonian H, define the vector field by $\iota_{X_H} \omega_s = -dH$. Nondegeneracy gives a unique smooth vector field; compactness gives its complete flow. Thus s may be any positive real value. There is no globally defined one-form with derivative ω_s, since the integral of an exact two-form on the closed sphere would vanish.

Instead use regular north and south potentials

$$\alpha_N = s(1 - \cos\theta) d\phi, \qquad \alpha_S = -s(1 + \cos\theta) d\phi. \tag{P2}$$

Each has derivative ω_s. On a simply connected part of the overlap,

$$\alpha_N - \alpha_S = 2s \, d\phi = d(2s\phi). \tag{P3}$$

The local actions $S_U = \int [\alpha_U(\dot{n}) - H(n)] dt$ differ by an endpoint term. Their fixed-endpoint variations agree, yielding the same global Hamiltonian equation. The azimuth is not single-valued on the entire overlap; going around it changes $2s\phi$ by $4\pi s$. This constant ambiguity does not change the equations. Classical variational consistency requires the local derivatives to agree, not a globally single-valued real action value.

As a concrete clock, take $H = -\Omega s \cos\theta$, where Ω has inverse time units. The equation gives $\dot{\phi} = \Omega$, $\dot{\theta} = 0$. Multiplying s and H by the same positive factor leaves this physical-time motion intact.

The additional phase premise

Now supply $K > 0$ with action units and ask for phase transport compatible with the local factors $\exp(iS_U/K)$. On the overlap the transition phase, up to convention for its inverse, is

$$g_{NS}(\phi) = \exp(i2s\phi/K).$$

This must be single-valued when ϕ advances by 2π. Hence

$$\exp(i4\pi s/K) = 1 \quad \Longleftrightarrow \quad \frac{2s}{K} \in \mathbb{Z}. \tag{P4}$$

The same condition follows by spanning a closed loop with a surface: two spanning surfaces differ by an integer number of spheres, so their geometric actions differ by $4\pi s$ times an integer. Equation (P4) makes the exponent independent of this choice. For open paths the transport is a map between endpoint phase fibres, rather than a gauge-independent scalar.

The extra premise is a U(P1) phase with curvature proportional to ω_s/K. Equation (P4) is its compatibility condition. The classical symplectic form alone supplies no such phase, no interference rule and no K. Prequantization also leaves further state-space and measurement choices open.

Common scaling, local caps and coupled spheres

Common rescaling. If (s,H,K) satisfies the classical and phase conditions, so does $(\eta s, \eta H, \eta K)$ for every $\eta > 0$. Both the Hamiltonian flow and the dimensionless phase remain unchanged. Thus even phase consistency fixes a ratio rather than an absolute dimensional constant.

Local contraction. With H=0, the geometric action around a latitude bounding a north-pole cap is

$$A(\theta_0) = 2\pi s(1 - \cos\theta_0) \longrightarrow 0 \quad (\theta_0 \downarrow 0). \tag{P5}$$

These are trial loops, not nonconstant solutions for H=0. They test local action resolution: quantization of the total sphere flux allows arbitrarily small nonzero cap actions. It does not impose a minimum polygonal cell or invalidate ordinary time subdivision. At fixed K and s>0, (P4) instead restricts the allowed total sphere area, giving $s \geq K/2$ in that family.

Two coupled spheres. Take the product with form $\omega_{s_1} \oplus \omega_{s_2}$ and any smooth interaction, for example $H = Jn_1 \cdot n_2$ with J in energy units. It defines complete classical dynamics for arbitrary $s_1, s_2 > 0$, including $s_2/s_1 = \sqrt{2}$. A common prequantization phase constant would require

$$2s_1/K = m_1, \qquad 2s_2/K = m_2, \qquad m_1, m_2 \in \mathbb{Z}_{>0}. \tag{P6}$$

Restriction to each sphere with the other held fixed proves these necessary conditions; product transition functions make them sufficient for this prequantization construction. Thus $s_2/s_1 = m_2/m_1$ must be rational. Classical coupling admits the irrational example, while a single-K phase bundle excludes it. The exclusion comes from the added phase structure, not from a failure of reciprocal Hamiltonian mechanics. Even a permitted rational ratio leaves infinitely many choices obtained by K replaced by K/n.

Source boundary for spin patching

F. Bonechi, A. S. Cattaneo and M. Zabzine, *Geometric quantization and non-perturbative Poisson sigma model*, Advances in Theoretical and Mathematical Physics **10**, 683–712 (2006), arXiv:math/0507223v3, explicitly attributes its integrality condition to requirements on a path integral. Coverage here: one discovery query and its arXiv abstract; no full-proof or exact sphere-formula audit. The local potentials, variational patching and tests (4)–(6) are written

out above. The source was not read for the exact sphere formulas; those calculations are the displayed written derivation, with no independent review or novelty claim.

Consequence for STATE

A connected reciprocal network fixes compatible mass ratios; canonical Newton derivations or mechanical rotation closure equate constituent algebra constants; a common spin phase imposes an integral ratio. These implications retain their different premises. Their common scale freedom, the classical zero branch and arbitrarily small local cap actions survive. Identification of a universal nonzero physical $\hbar$ still needs independent physical preparation, state/dynamics and calibration laws.